1. Objective

What was the task about?

In this task, I learned how to gather information about a target domain without directly interacting with it. I understood how WHOIS, DNS lookups, DNSDumpster, crt.sh, and Shodan help identify domain details, DNS records, subdomains, and publicly exposed services.

2. Concept / Understanding

What did you learn?

I learned how passive reconnaissance helps gather information about a domain without directly interacting with the target. I understood how WHOIS provides domain registration details, DNS lookups reveal different DNS records, DNSDumpster and crt.sh help discover subdomains, and Shodan identifies publicly exposed services. I also learned the importance of using these techniques ethically and within an authorised scope.

3. Practical Work

What did you perform? Write only what you actually did.

1. I studied the concept of passive reconnaissance and learned how to gather information without directly interacting with the target.

2. I learned how WHOIS helps identify domain registration details, registrar information, registration dates, and name servers.

3. I studied the WHOIS command used for domain information gathering.

Command: whois tryhackme.com

4. I learned about DNS records and how they provide information about a domain's IP addresses and services.

5. I studied how to use nslookup to find IPv4 address records.

Command: nslookup -type=A tryhackme.com

6. I learned how to identify mail servers using MX records.

Command: nslookup -type=MX tryhackme.com 1.1.1.1

7. I studied how TXT records contain information related to SPF, DMARC, and domain verification.

Command: nslookup -type=TXT tryhackme.com

8. I learned how to use the dig command as an alternative for DNS lookups.

Command: dig tryhackme.com A

9. I studied how to query MX records using a specific public DNS resolver.

Command: dig @1.1.1.1 tryhackme.com MX

10. I learned how to retrieve TXT records using the dig command.

Command: dig tryhackme.com TXT

11. I explored DNSDumpster to understand how DNS information can be collected and displayed in a graphical format.

Website: https://dnsdumpster.com/

12. I learned how Certificate Transparency logs can help discover subdomains associated with a domain.

Website: https://crt.sh/

13. I studied the wildcard search format used to find certificates associated with subdomains.

Search example: %.tryhackme.com

14. I learned how Shodan provides information about publicly exposed internet-connected devices and services.

Website: https://www.shodan.io/

15. I studied how Shodan can display service banners, open ports, and hosting information.

16. I learned how public DNS resolvers, such as 1.1.1.1, can be specified when performing DNS queries.

17. I understood that IP addresses and DNS records may change over time, so reconnaissance results should be checked carefully.

18. I learned how publicly available information can reveal forgotten subdomains, outdated services, and possible security misconfigurations.

19. I studied how defenders can monitor Shodan/Censys alerts, Certificate Transparency logs, and DNS changes to identify potential security risks.

20. I reviewed the importance of conducting reconnaissance ethically and ensuring that all security activities remain within the authorised scope.

4. Tools Used

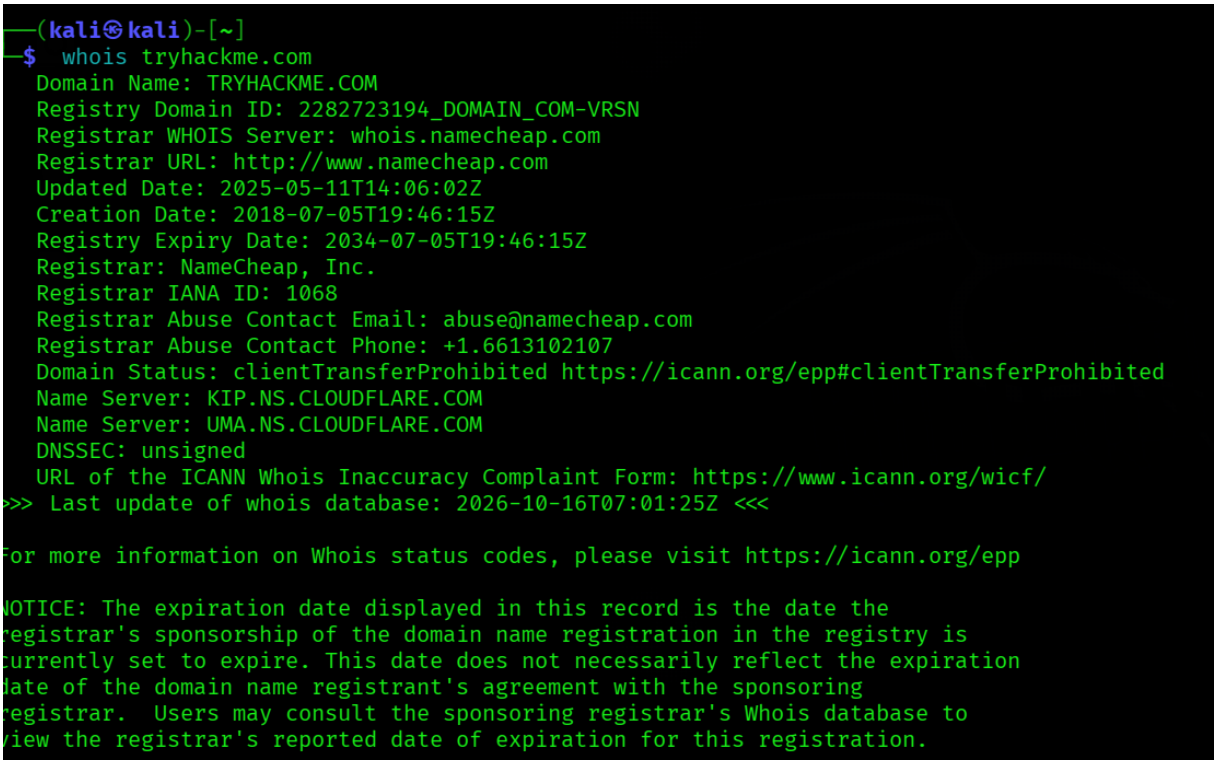
Tools used during the task.

Tool	Purpose
WHOIS	To find domain registration details, registrar information, and name servers.
nslookup	To retrieve DNS records such as A, MX, and TXT records.
dig	To perform DNS lookups and query specific DNS record types.
DNSDumpster	To gather DNS information and discover subdomains.

Tool	Purpose
Shodan.io	To identify publicly exposed devices, services, ports, and service banners.

5. Evidence

Relevant screenshots and results. Real screenshots only.





```
┌──(kali㉿kali)-[~]
└─$ nslookup -type=MX tryhackme.com
Server:         192.168.233.2
Address:        192.168.233.2#53

Non-authoritative answer:
tryhackme.com   mail exchanger = 5 alt2.aspmx.l.google.com.
tryhackme.com   mail exchanger = 1 aspmx.l.google.com.
tryhackme.com   mail exchanger = 10 alt3.aspmx.l.google.com.
tryhackme.com   mail exchanger = 10 alt4.aspmx.l.google.com.
tryhackme.com   mail exchanger = 5 alt1.aspmx.l.google.com.

Authoritative answers can be found from:
```

```
┌──(kali㉿kali)-[~]
└─$ dig @1.1.1.1 tryhackme.com MX

; <<>> DiG 9.20.26-1-Debian <<>> @1.1.1.1 tryhackme.com MX
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 48428
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;tryhackme.com.                 IN      MX

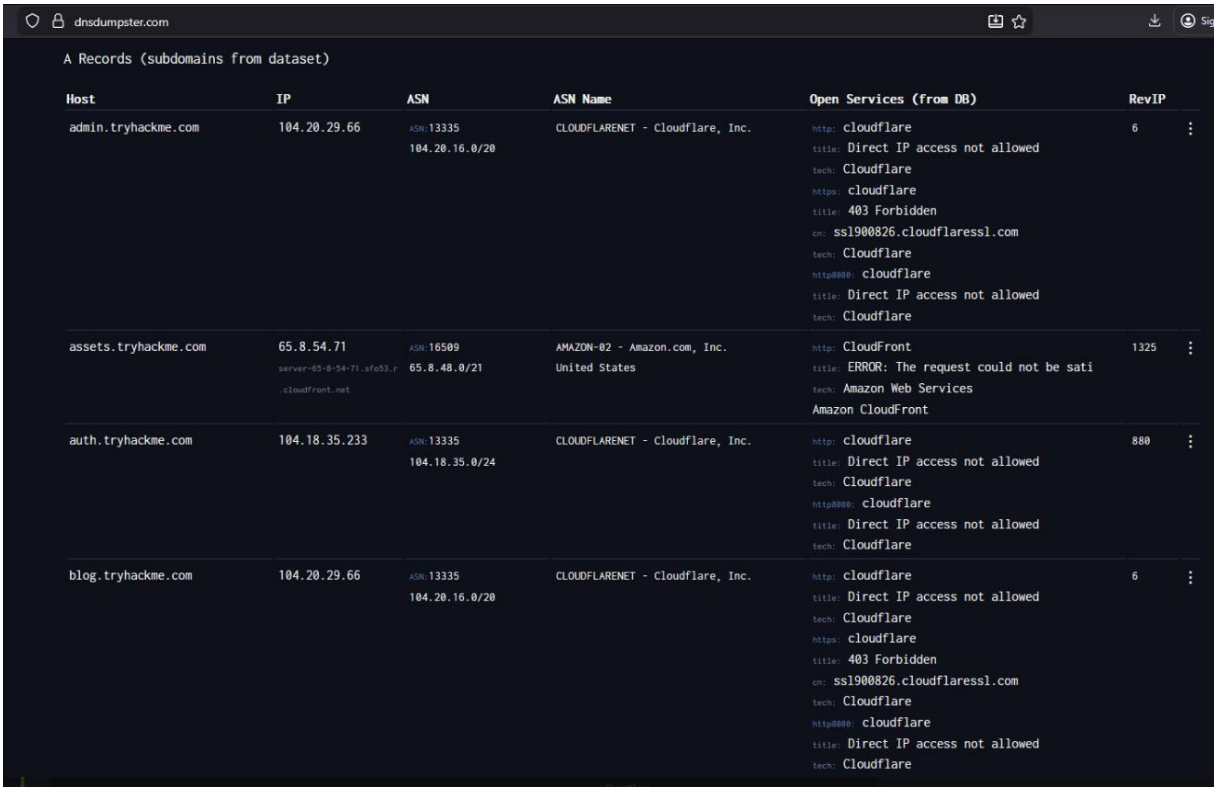
;; ANSWER SECTION:
tryhackme.com.          300     IN      MX      1 aspmx.l.google.com.
tryhackme.com.          300     IN      MX      10 alt3.aspmx.l.google.com.
tryhackme.com.          300     IN      MX      10 alt4.aspmx.l.google.com.
tryhackme.com.          300     IN      MX      5 alt1.aspmx.l.google.com.
tryhackme.com.          300     IN      MX      5 alt2.aspmx.l.google.com.

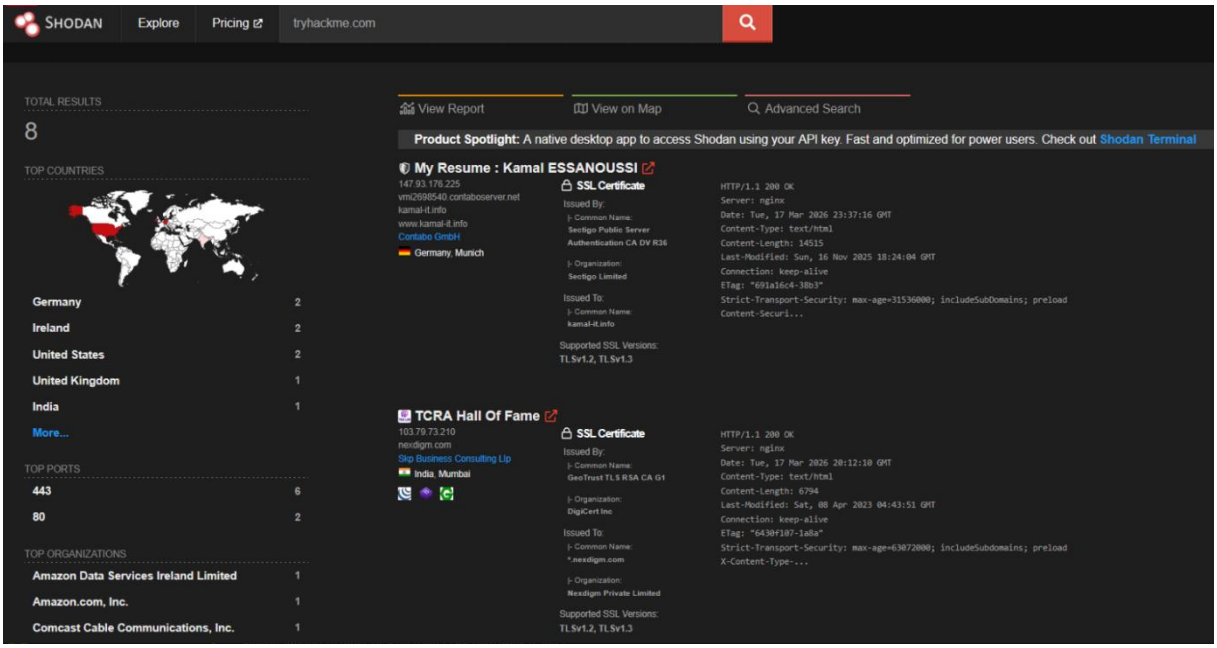
;; Query time: 27 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Fri Oct 16 03:03:24 EDT 2026
;; MSG SIZE  rcvd: 157
```

```
┌──(kali㉿kali)-[~]
└─$ nslookup -type=A tryhackme.com 1.1.1.1
Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
Name:   tryhackme.com
Address: 64.239.109.193
Name:   tryhackme.com
Address: 64.239.123.193
```


tryhackme.com
MX
DNS
5 alt2.aspmx.l.google.com
1 aspmx.l.google.com
10 alt3.aspmx.l.google.com
10 alt4.aspmx.l.google.com
5 alt1.aspmx.l.google.com
kip.ns.cloudflare.com
uma.ns.cloudflare.com
auth.tryhackme.com
blog.tryhackme.com
business.tryhackme.com
docs.tryhackme.com
admin.tryhackme.com
insights-proxy-worker.tryhackme.com
guacaworker.tryhackme.com
172.253.116.26
net: 172.253.0.0/16
192.178.155.27
net: 192.178.0.0/15
192.178.223.26
net: 192.178.0.0/15
173.194.76.27
net: 173.194.0.0/16
108.177.123.26
net: 108.177.0.0/17
15169
GOOGLE - Google LLC
173.245.59.128
net: 173.245.59.0/24
173.245.58.146
net: 173.245.58.0/24
104.18.35.233
net: 104.18.35.0/24
104.20.29.66
net: 104.20.16.0/20
172.66.164.239
net: 172.66.160.0/20
23.227.38.74
net: 23.227.38.0/23
13335
CLOUDFLARENET - Cloudflare, Inc.


dnsdumpster.com
A Records (subdomains from dataset)
Host
IP
ASN
ASN Name
Open Services (from DB)
RevIP
admin.tryhackme.com
104.20.29.66
ASN:13335
104.20.16.0/20
CLOUDFLARENET - Cloudflare, Inc.
http: cloudflare
title: Direct IP access not allowed
tech: Cloudflare
https: cloudflare
title: 403 Forbidden
cn: ssl900826.cloudflaressl.com
tech: Cloudflare
http8080: cloudflare
title: Direct IP access not allowed
tech: Cloudflare
6
assets.tryhackme.com
65.8.54.71
server-65-8-54-71.sfo53.r.cloudfront.net
ASN:16509
65.8.48.0/21
AMAZON-02 - Amazon.com, Inc.
United States
http: CloudFront
title: ERROR: The request could not be sati
tech: Amazon Web Services
Amazon CloudFront
1325
auth.tryhackme.com
104.18.35.233
ASN:13335
104.18.35.0/24
CLOUDFLARENET - Cloudflare, Inc.
http: cloudflare
title: Direct IP access not allowed
tech: Cloudflare
http8080: cloudflare
title: Direct IP access not allowed
tech: Cloudflare
880
blog.tryhackme.com
104.20.29.66
ASN:13335
104.20.16.0/20
CLOUDFLARENET - Cloudflare, Inc.
http: cloudflare
title: Direct IP access not allowed
tech: Cloudflare
https: cloudflare
title: 403 Forbidden
cn: ssl900826.cloudflaressl.com
tech: Cloudflare
http8080: cloudflare
title: Direct IP access not allowed
tech: Cloudflare
6



6. Result

What did you achieve or discover?

I successfully learned how passive reconnaissance works and how to gather publicly available information about a domain using WHOIS, DNS lookups, DNSDumpster, crt.sh, and Shodan. I understood how these tools help identify domain details, DNS records, subdomains, and publicly exposed services. I also learned the importance of performing reconnaissance ethically and staying within the authorised scope.

7. Key Learnings

What did you learn from the task?

- I learned how passive reconnaissance helps gather publicly available information about a target without directly interacting with it.
- I learned how WHOIS and DNS lookup commands help identify domain details and DNS records.
- I understood how DNSDumpster and crt.sh help discover subdomains.
- I learned how Shodan identifies publicly exposed devices, ports, and services.
- I understood the importance of DNS record types such as A, AAAA, MX, and TXT.
- I learned how to use commands like whois, nslookup, and dig for basic information gathering.
- I understood how passive reconnaissance can reveal forgotten subdomains and potential security misconfigurations.
- I learned the importance of conducting reconnaissance ethically and within an authorised scope.